



Monday Sep-08-2025

Flowserve Facility:

Flowserve Nordstrom / Valbart Valves
1511 Jefferson St.
Sulphur Springs, TX 75482

To:

cmtrs@flowserve.com

CERTIFICATE OF CONFORMANCE

SERIAL #	HEAT #	PO #	Description	Valve Type	Class	End Connection
61-698884	698884	662668300	10" 2349 A SP	PLUG	900	FLG
61-698884	3224674AB0	662668300	10" 2349 A SP	PLUG	900	FLG
61-698884	AA2716	662668300	10" 2349 A SP	PLUG	900	FLG
61-698885	698885	662668300	10" 2349 A SP	PLUG	900	FLG
61-698885	3224674AB0	662668300	10" 2349 A SP	PLUG	900	FLG
61-698885	AA2717	662668300	10" 2349 A SP	PLUG	900	FLG
61-698886	698886	662668300	10" 2349 A SP	PLUG	900	FLG
61-698886	3224674AB0	662668300	10" 2349 A SP	PLUG	900	FLG
61-698886	AA2716	662668300	10" 2349 A SP	PLUG	900	FLG
61-698887	698887	662668300	10" 2349 A SP	PLUG	900	FLG
61-698887	3224674AB0	662668300	10" 2349 A SP	PLUG	900	FLG
61-698887	V7557	662668300	10" 2349 A SP	PLUG	900	FLG

We hereby certify that all the subject valves/parts/sealant have been manufactured in conformance with the applicable requirements of Flowserve Sulphur Springs Operation (FSSO) specifications and that the valve/parts/sealant materials for component parts meet the applicable requirements of FSSO drawings and ASTM specifications, ASME B16.34 (if Applicable), API Q1 10th edition, and API 6D 25th edition (if applicable to the API monogram). Flowserve Dynamic Balanced Valves are provided with a Pressure balance feature unless specifically identified in the Valve description. All valves/parts/sealant meet all the requirements of the customer purchase order.

Documentation Technician
Justin McKee



SULPHUR SPRINGS OPERATION

PLUG VALVE TEST RECORD

SIZE: 10	PRESSURE: 900
FIGURE NUMBER: H2349 A SP	
B/M NO.: F01700482	
ASSEMBLY ORDER NO.: H3600623	

VALVE TESTED PER RMP: 61196	
E.I DRAWING NO.:	REV: 19
	21700482

API SERIAL NO.:	61-698885
MMN SERIAL NO.:	B96137
BODY SERIAL NO.:	
BODY HEAT NO.:	AA2717
BODY RADIOGRAPH NO.:	X
COVER HEAT NO.:	3224674AB0
STEM JOB NO.:	X
PLUG JOB NO.:	X
TYPE OF OPERATOR:	OHM-TECH
SEALANT:	950

	OPERATION	SHELL	SEAT
TEST PRESSURES, PSIG	2250	3400	2445
TEST DURATION TIME, MIN.		15	5

TEST RESULTS		ACCEPTANCE
SHELL HYDROSTATIC TEST	PASSED: ☒ FAILED: ☐	NO LEAKAGE
PRESS. GAUGE RANGE & ID. No.	HYDRO-2	
DIFFERENTIAL PRESSURE TORQUE	AA= SEAT 19 FT-LBS.	MAX: 27
	AB= SEAT 19 FT-LBS.	
SEAT HYDROSTATIC	AA= SEAT PASSED: ☒ FAILED: ☐	NO LEAKAGE
	AB= SEAT PASSED: ☒ FAILED: ☐	
PRESS. GAUGE RANGE & ID. No.	HYDRO-2	
SEAT AIR (IF SPECIFIED)	AA= SEAT PASSED: ☐ FAILED: ☐	NO LEAKAGE
	AB= SEAT PASSED: ☐ FAILED: ☐	
PRESS. GAUGE RANGE & ID. No.		
SHELL AIR TEST	PASSED: ☐ FAILED: ☐	

SPECIAL TESTS PER ENGINEERING INSTRUCTION

FINAL INSPECTION TORQUE TESTS - ZERO PRESSURE			
INITIAL BREAK MAX.	FT-LBS.	RUNNING	6 FT-LBS.
ACTUAL	FT-LBS.	ACTUAL	4 FT-LBS.

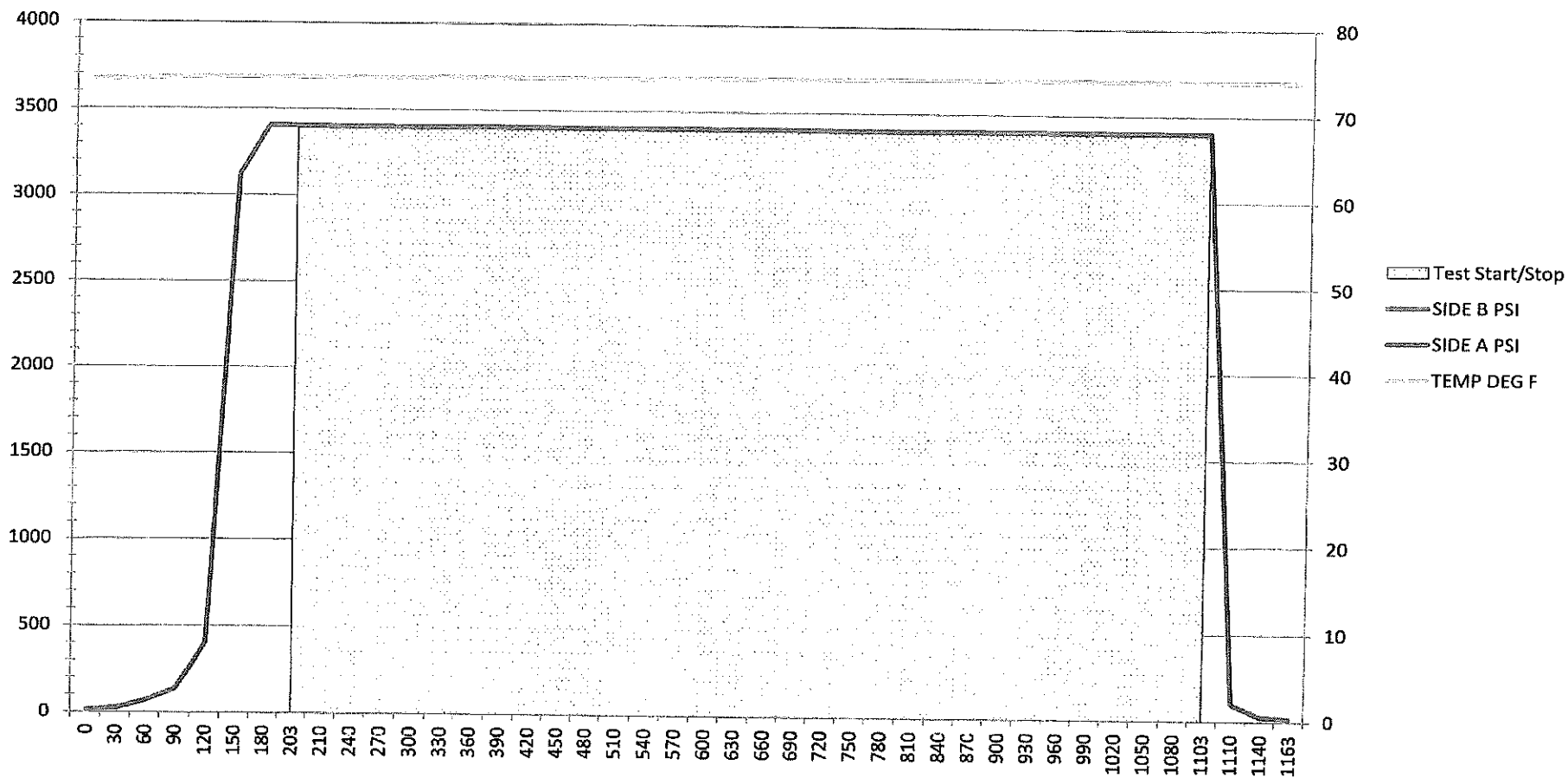
DATE TESTED:	TESTED BY:	ACCEPTED BY:	CUSTOMER WITNESS BY:
9/5/25	J POTTS	MM	<i>Roger J. Potts</i>

ENG102
REV. 8

Roger J. Potts 9-5-25

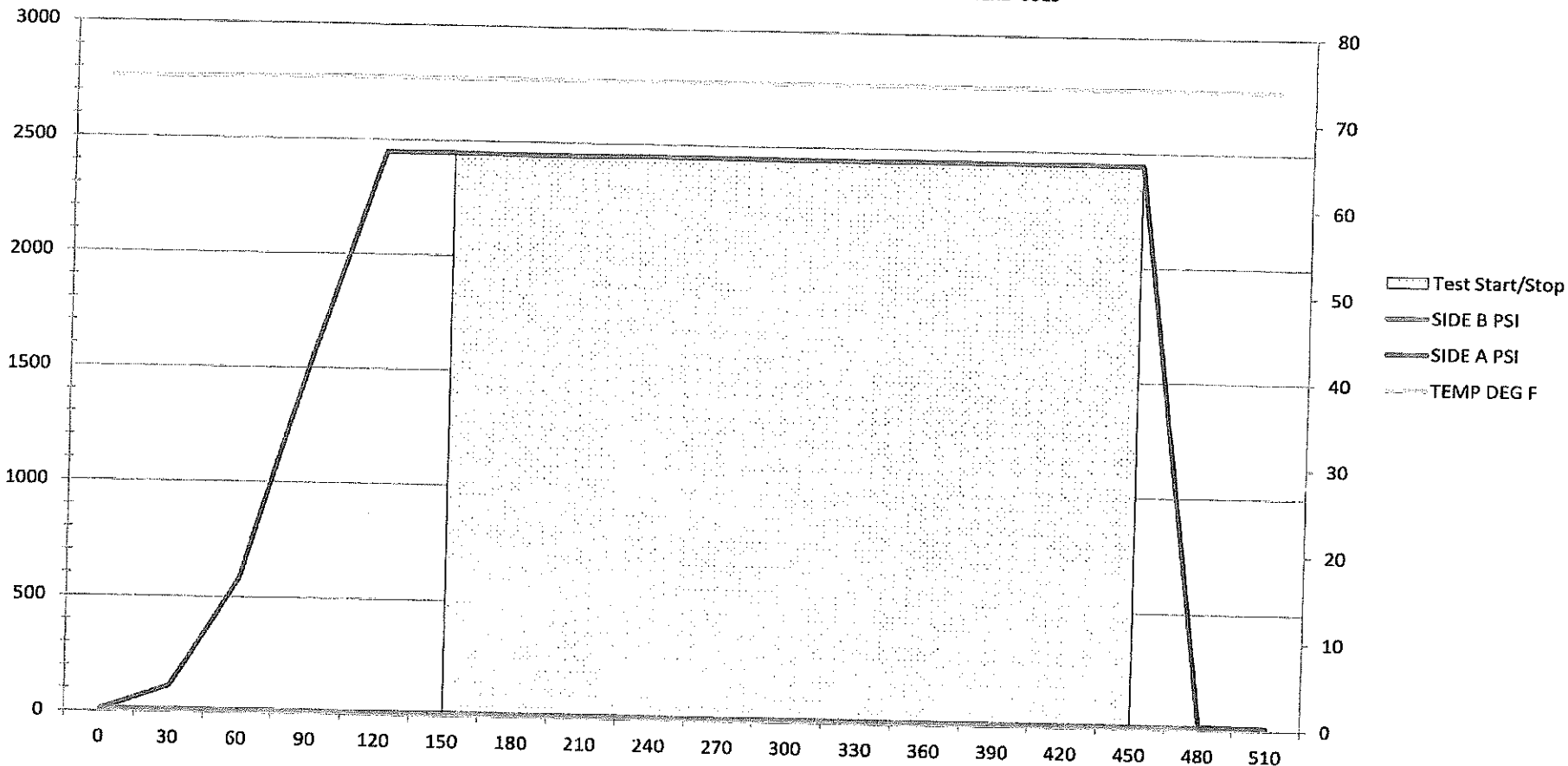
FILE NAME : HYDRO 2(SHELL) (083043) (09052025)
JOB# : H3600623
BODY HEAT# : AA2717
COVER HEAT# : NA
BODY S/N : 896137
EMPLOYEE NAME : J POTTS
API S/N : 61 698885
MISC 1 : FLOWERVE
MISC 2 : 10 IN H2349 A SP

CHART START TIME (HHMMSS) : 81113
CHART START DATE (MMDDYYYY) 9052025
TEST START ELAPSED SECONDS : 203
TEST END ELAPSED SECONDS : 1103
CHART RECORD END ELAPSED SECONDS : 1163
CHART MAX LENGTH HOURS : 24
HIGH PRESSURE TRANSMITTERS WERE USED



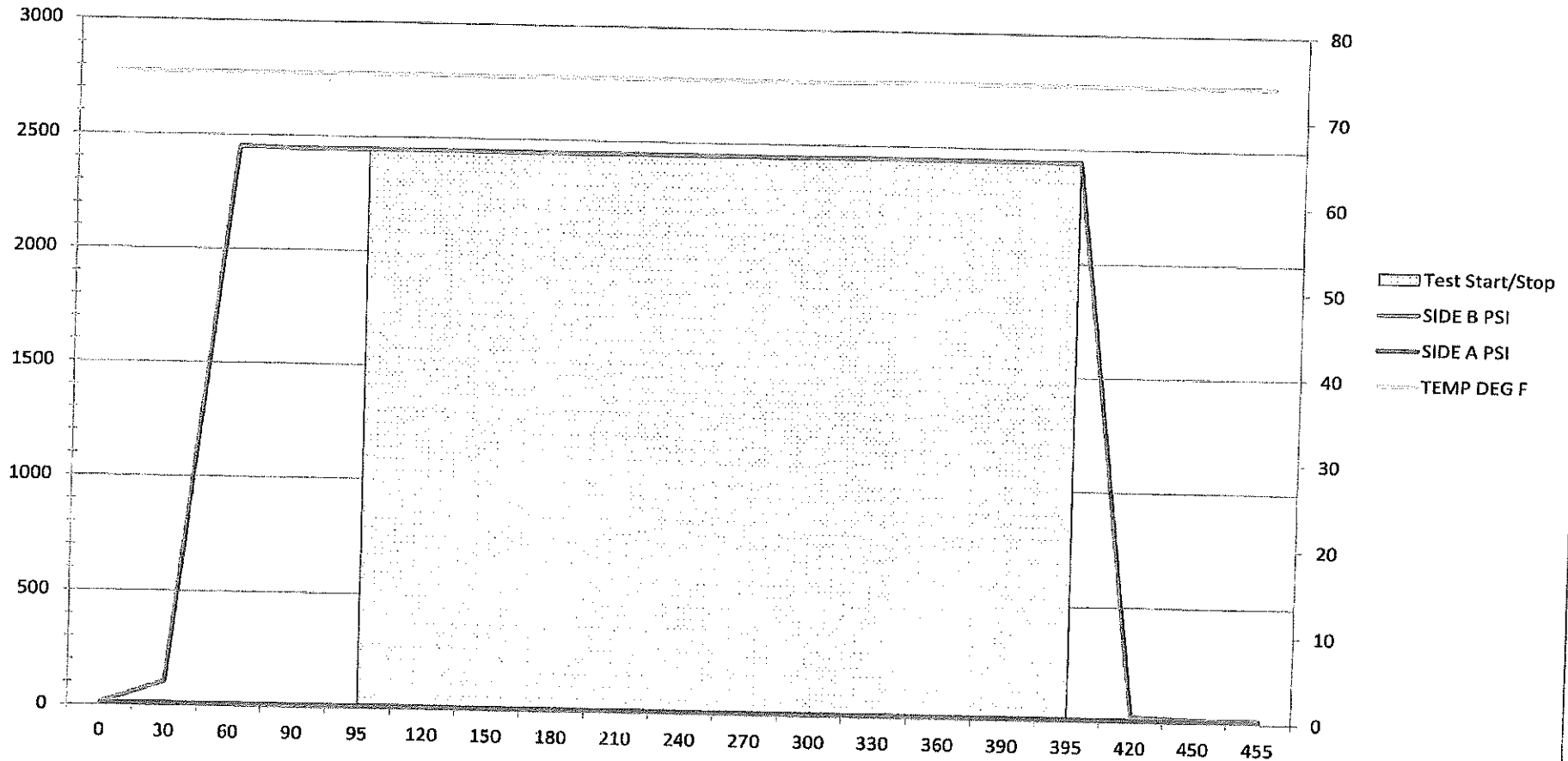
FILE NAME : HYDRO 2(A_HST)(084202)(09052025)
JOB# : H3600623
BODY HEAT# : AA2717
COVER HEAT# : NA
BODY S/N : 896137
EMPLOYEE NAME : J POTTS
API S/N : 61 698885
MISC 1 : FLOWERVE
MISC 2 : 10 IN H2349 A SP

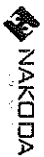
CHART START TIME (HHMMSS) : 83303
CHART START DATE (MMDDYYYY) 9052025
TEST START ELAPSED SECONDS : 150
TEST END ELAPSED SECONDS : 450
CHART RECORD END ELAPSED SECONDS : 510
CHART MAX LENGTH HOURS : 24
HIGH PRESSURE TRANSMITTERS WERE USED



FILE NAME : HYDRO 2(B_HST)(085050)(09052025)
JOB# : H3600623
BODY HEAT# : AA2717
COVER HEAT# : NA
BODY S/N : B96137
EMPLOYEE NAME : J POTTS
API S/N : 61 698885
MISC 1 : FLOWSERVE
MISC 2 : 10 IN H2349 A SP

CHART START TIME (HHMMSS) : 84233
CHART START DATE (MMDDYYYY) 9052025
TEST START ELAPSED SECONDS : 95
TEST END ELAPSED SECONDS : 395
CHART RECORD END ELAPSED SECONDS : 455
CHART MAX LENGTH HOURS : 24
HIGH PRESSURE TRANSMITTERS WERE USED





NAKODA ENGINEERS



TEST CERTIFICATE AS PER EN 10204 3.1

TC No.:	0454	Date:	23/05/2025
Customer:	FLOWSERVE INDIA CONTROLS PRIVATE LIMITED	Purchase order ref:	10124208
Material Specification & year of edition	ASTM A516 Gr.70	Invoice/Delivery challan ref:	NEB/0454/25-26
		Date:	23-05-2025

PART DETAILS

Item Code	Size/Description	No. of Plates	Plate no.	Heat no.	Mill supplier	Mill TC reference
N122327	COVER MCD 10" CL.900 GR.70 FIG NO.2349	5	AB0	3224674	JINDAL STEEL & POWER LIMITED	0000006169

CHEMICAL ANALYSIS %

Specification value	C	S	P	Mn	Si	Cu	Ni	Cr	Mo	V	Co	Ti	CE
Minimum	0.000	0.000	0.000	0.390	0.130	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Maximum	0.300	0.025	0.025	1.500	0.450	0.430	0.430	0.340	0.130	0.040	0.030	0.040	0.430
Mill TC value	0.160	0.004	0.021	1.380	0.270	0.035	0.011	0.098	0.006	0.001	0.016	0.013	0.410
Tested value	0.199	0.010	0.024	1.134	0.374	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.400

Chemical Test done at OMEGA INSPECTION & ANALYTICAL LABORATORY Lab TC ref: TC NO:OAL/2025/04/00631 DATE:09.04.2025

MECHANICAL ANALYSIS

Specification value	Yield Strength @ 0.2% offset method	Tensile Strength	Elongation in 2"	RA	Hardness (BHN)	Impact at -50 Deg F Min: 27 Joules Single specimen : Avg : 27 Joules			
Minimum	38 Ksi (260 Mpa)	70 Ksi (485 Mpa)	21%						
Maximum	-	90 Ksi (620 Mpa)	-		233 (99 HRB)	1	2	3	Average
Mill TC value	54.82 Ksi (378 Mpa)	76.73 Ksi (529 Mpa)	35%		156	116	120	124	120
Tested value	43.00 Ksi (297 Mpa)	76.00 Ksi (527 Mpa)	28%		81, 78, 79 (HREW)				

Mechanical Test done at OMEGA INSPECTION & ANALYTICAL LABORATORY TC NO:OAL/2025/04/00631 DATE:09.04.2025

Heat Treatment condition: Normalized at 910±10°C, Soaking time 1min/lmm or 60min whichever is more & air cooled.

MARKING

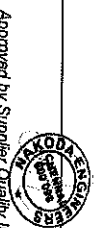
The plates are marked as "A516 GR.70", "N122327" (3224674AB0).

The Material conforms to ASTM A516-17 Gr.70, NACE MR0175-2021/ISO 15156-1:2020 and Flowserve Specification: RMC25024-REV.08100.

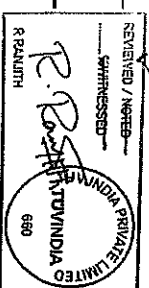
We hereby certify that the items mentioned above have been Visually, Dimensionally Inspected, Tested and found to be in accordance with Specification requirements.



Supplier Quality (RAGUL.S)



Approved by Supplier Quality Incharge (RAGUL.S)



R.RAJATH

INSPECTION CERTIFICATE
AS PER EN 10204:2004 Type 3.1Customer : FLOWSERVE INDIA CONTROLS PVT LTD (NORDSTROM)
KANCHIPURAM 603 209T.C. No : FD/AA2717/25
T.C.Date : 30/04/2025

P.Order : 10124314 Dt. 22/03/2025

Rev No : Dt:-

Heat No : AA2717
Pour Date : 16/04/2025

Specification : ASTM A216/A216M Gr WCC-2021 [UNS J02503]

CASTING PROCESSED BY ELECTRIC INDUCTION MELTING
FOUNDRY IDENTIFICATION : VSF

CHEMICAL COMPOSITION % (ASTM A751-2021)

Element%	C	Mn	Si	P	S	Cr	Ni	Mo	Cu	V	#CE	*TRE	N	Al
Minimum	---	---	---	---	---	---	---	---	---	---	---	---	---	---
Maximum	0.230	1.20	0.600	0.025	0.020	0.500	0.500	0.200	0.300	0.030	0.420	1.00	0.015	0.080
Actual Value	0.210	1.02	0.440	0.021	0.009	0.100	0.090	0.020	0.010	0.008	0.411	0.228	0.009	0.054

HEAT TREATMENT : Normalizing - Heated to 920°C soaked for 4.00 hrs and air cooled

MECHANICAL PROPERTIES (ASTM A370-2024a)(Room temperature:22°C)

Requirements	Yield Strength		Tensile Strength		% Elongation	% Reduction	Hardness		Bend Test	Impact Test in (J)	
	Kg/1mm ²	Ksi	Kg/1mm ²	Ksi	GL:50 mm	In Area	HBW	AT90 ° D=11		Size : (10mm X 10mm X 55mm	
Minimum	28.04	40	49.46	70	22	35	---	---	-		
Maximum	---	---	66.79	95	---	---	233	-	1	2	3 Avg
Actual Value	35.49	50	55.88	79	35.36	60.94	154,155,154 AVG:154	Passed	--	--	--

ITEM DESCRIPTION

S.L.No.	Description	Drawing No.	Part No.	Pour S.L. No.	Quantity
1	10" CL 900 FL BODY (F.NO.2349)	122717 / REV.15	IN122717WCC	1, 2	2
REMOVED / NOTED - SUBMITTED P. ROSE INDIA PRIVATE LIMITED R. RAJATH 660					

Remarks : Visual inspection as per MSS SP-55-2011, level of Type II-XII a found Satisfactory.

No surface irregularities to Type I found, surface irregularities to Type II-XII do not exceed example "a"

#CE= C+Mn/6+Cr+Mo+V/5+Ni+Cu/15 & *TRE = Cr+Ni+Mo+Cu+V.

This Material conforms to NACE MR0175-2021/ ISO15156-2-2020, FTDCC01R8 & Material Spec No:RMC90012-Rev15 Requirements.

Carbon steel fully killed & fine grained condition. The castings are free from radioactive contamination / radiation.

QA system have undergone Specific Assessment of Materials according to Annex-1, Clause 4.3 of PED/2014/68/ EU Certified By TUV Nord Vide Certificate No:0045/2021WZ/1291/24/Rev.1 Valid till May 2027.

We Conform that the above material was manufactured, sampled, tested and inspected in accordance with the drawings, customer order requirements, material specification ASTM A216/A216M Gr WCC-2021 [UNS J02503] and was found to meet the requirements & released for dispatch.

S. Balasubramanian
Sr. Chemist
Prepared byM. Jeyakumar
Manager QA/QC
Approved by